Requirement Traceability Matrix_Project_Healthcare_Version1_Dt.24/11/2025

Requirement_id	Requirement_description	Priority_level(MoSCoW)	Stakeholder(s)	Project Objective	Related Data File	Status
FR001	**Automated Booking and Scheduling System** Automated Booking and Scheduling system that enables patients and administrative staff to efficiently book and schedule appointments based on real-time resource availability, minimizing overbooking	Must Have	Patients & Administrative Staff	To efficiently and easily take appointments and schedule patients and doctors thus avoiding overbooking of resources and last minute cancellation	Appointment & Feedback Data	In-progress
FR002	**Integrated Hospital Information System (HIS)** The HIS must maintain accurate patient records, synchronize data across departments, and ensure seamless information flow	Must Have	Admin & IT teams	To ensure proper data synchronisation between different systems so that there is visibility in resource availability and also proper and real time data is available for efficient service	Resource Data	To Start
FR003	**Real-Time Communication Module** The system must provide timely communication and notifications (via email/SMS) to patients and staff regarding appointment confirmations, cancellations, or reschedules, ensuring updates are delivered to the appropriate recipients without delay	Should Have	Admin, IT teams, Nurses, Doctors and Patients	To ensure that the changes in appointments, schedules are properly communicated to the responsible personnel to avoid miscommunication/ plan their resource/travel such that, they are efficiently used and can better prepare themselves	All data files	In Progress

NFR001	**Usability & Performance** •The booking and scheduling interface must be intuitive and user-friendly with a clean layout. •The system should maintain **peak operational efficiency** with **≤2% error rate** and **page load times ≤3 seconds** during high-traffic periods.	Should Have	IT teams	To enable users to book easily and effectively without any issues-simple and clean layout	Feedback Data	Completed
NFR002	**Reliability & Data Synchronization** •All records must remain synchronized across modules in real-time. •Any data-sync failure or delay must trigger an automated warning or alert to the IT administrator. •The system should be capable of handling concurrent data requests without performance degradation.	Must Have	IT Team, Admins	To ensure the data records are properly synchronised such that there is no double booking or availability related issues, transparency in availability and booking and live data tracking.	Resource Data	In Progress

NFR003	**Notification Accuracy & Timeliness** •Any change in appointments (reschedule, delay, or cancellation) must trigger an automated notification to all affected stakeholders within **1 minute** of the update. •If a change occurs too close to the appointment time (e.g., within 15 minutes), the system must display a **priority warning** to the scheduling staff.	Could Have	IT teams	To ensure last minute changes are not encouraged and immediate notification of changes in schedules to the respective personnel	Resource data, appointment data	To Start